Test 1 - Low-Price Charging Test

Purpose

The purpose of this test is to verify that the EMS follows the correct **low-price charging logic** when real load data and real API price data are used.

The test should confirm that:

Condition	Expected scenario
LOW price + no PV	**S1**
LOW price + PV + battery not full	**S2**
LOW price + PV + battery full	**S3**

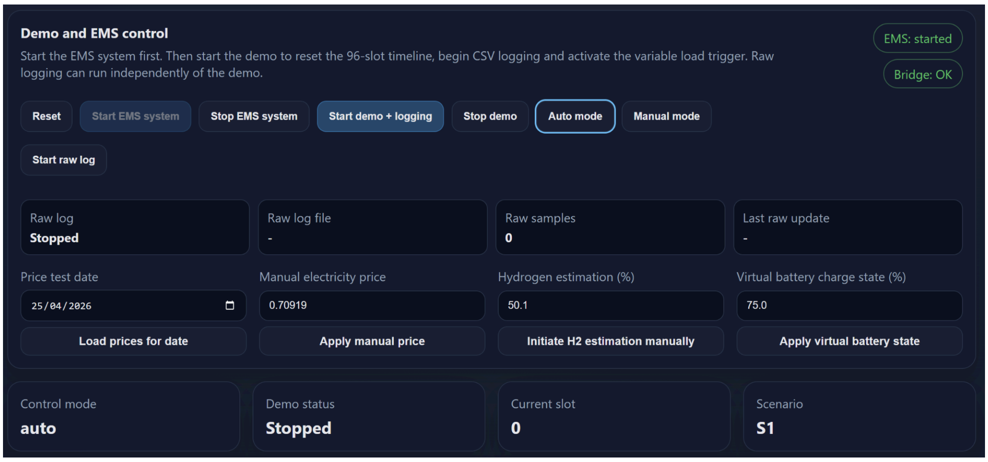
The main transition to observe is:

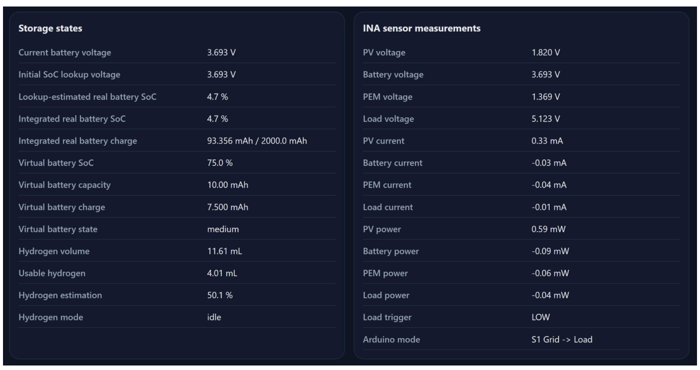
S2 → **S3**, when the virtual Battery SoC reaches the “full” threshold.

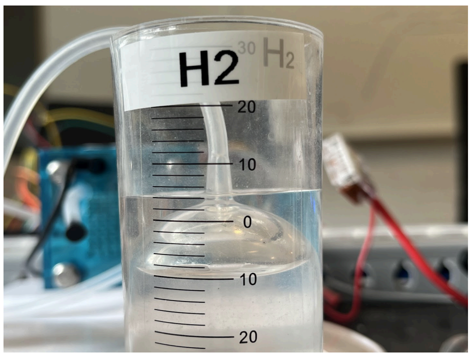
Initial setup

Using Scripts:
AppLab: ems_base_estimation_logger_v5
Arduino IDE: ControllableLoad_V4.1_triggerAndData

Item	Value
Test duration	24 minutes
Simulated duration	24 hours
Price data	Real API day with LOW prices
Load profile	Active, 15–75 mW
Initial Battery SoC	75%
Initial PEM SoC	50.1%
PV/lamp at start	Off then on at 6:00 to 18:30
CSV logging	Enabled







Hydrogen Niveau ved start af test (50% = 4mL)

Test procedure

1. Load the selected real API price day with LOW prices.
2. Set the initial Battery virtual SoC.
3. Set the initial PEM virtual SoC.
4. Start the load profile.
5. Start CSV logging.
6. Start the EMS test with the lamp turned off.
7. Turn the lamp on when the simulated daytime period begins (06:30).
8. Wait for the battery to reach the upper SoC threshold.
9. Observe whether the EMS switches from **S2** to **S3** when the battery becomes full.
10. Lamp is turned off again at 18:30
11. Test is allowed to finish
12. Stop the test after 24 minutes.
13. Save the CSV file and screenshots.

CSV file

Item	Entry
CSV filename	/app/python/logs/ems_demo_20260512_122830.csv
Saved location	Bachelor_repo/Bachelor/data/EMS_system_test/TEST1

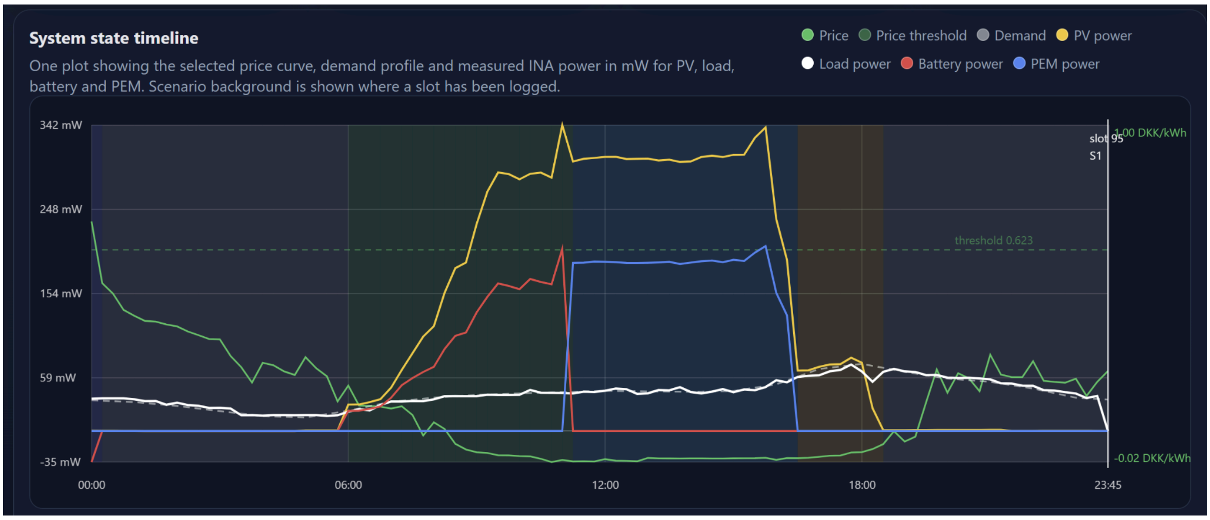
Notes

Time / timestep	Observation
00:15	S5 Battery to Load, Because Prices were high in the first slot
06:00 - 11:00	Lamp is Switched on and gradually lowered
06:00	S2 Grid to Load, PV to Battery
11:30	S3 Grid to Load, PV to PEM
16:00 - 18:30	Lamp is gradually raised again
16:30	S4 PV to Load (after PEM is fully charged)
18:30	S1 Grid to Load (after PV is turned off again)

Result summary

Question	Answer
Did the system start in S1 when PV was unavailable?	The system started in Batt to load. But this was because of a high Price slot. Otherwise S1 was used until PV was available.
Did the system switch to S2 when PV became available?	Yes switch happened at 06:00
Did Battery SoC increase during S2?	Yes the battery SoC increased from 75% to 90%
Did the EMS switch to S3 when Battery SoC reached the full threshold?	Yes the system succesfully switched to S3 when Battery SoC reached 90%

Did PEM SoC increase during S3?	PEM charged as expected from 50.1% to 100% After this the system switched back to PV to Load
Any unexpected scenario changes?	
Overall result	Pass

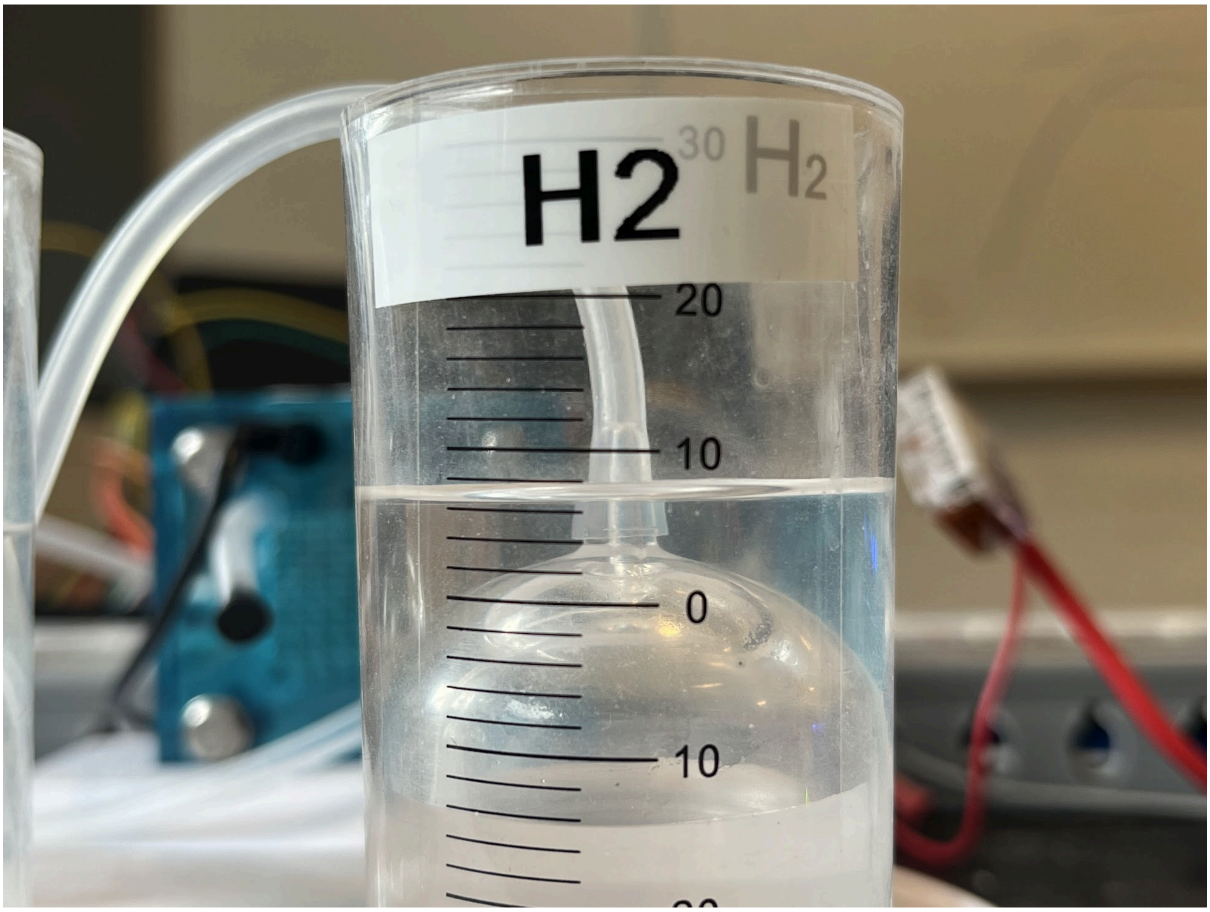


Storage states

Current battery voltage	3.692 V
Initial SoC lookup voltage	3.693 V
Lookup-estimated real battery SoC	4.6 %
Integrated real battery SoC	4.8 %
Integrated real battery charge	95.714 mAh / 2000.0 mAh
Virtual battery SoC	90.0 %
Virtual battery capacity	10.00 mAh
Virtual battery charge	9.005 mAh
Virtual battery state	control_full
Hydrogen volume	15.60 mL
Usable hydrogen	8.00 mL
Hydrogen estimation	100.0 %
Hydrogen mode	idle

INA sensor measurements

PV voltage	2.850 V
Battery voltage	3.692 V
PEM voltage	1.302 V
Load voltage	5.130 V
PV current	0.00 mA
Battery current	-0.03 mA
PEM current	-0.04 mA
Load current	-0.01 mA
PV power	0.00 mW
Battery power	-0.09 mW
PEM power	-0.06 mW
Load power	-0.04 mW
Load trigger	LOW
Arduino mode	S1 Grid -> Load



Hydrogen Niveau ved slutning af test (100% = 8mL)